

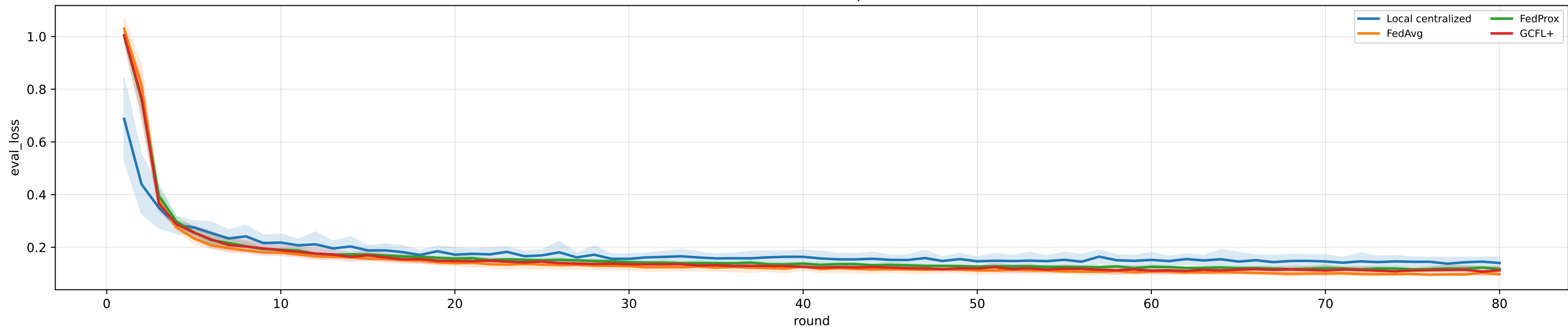
Controlled task-specialization level: Mid | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

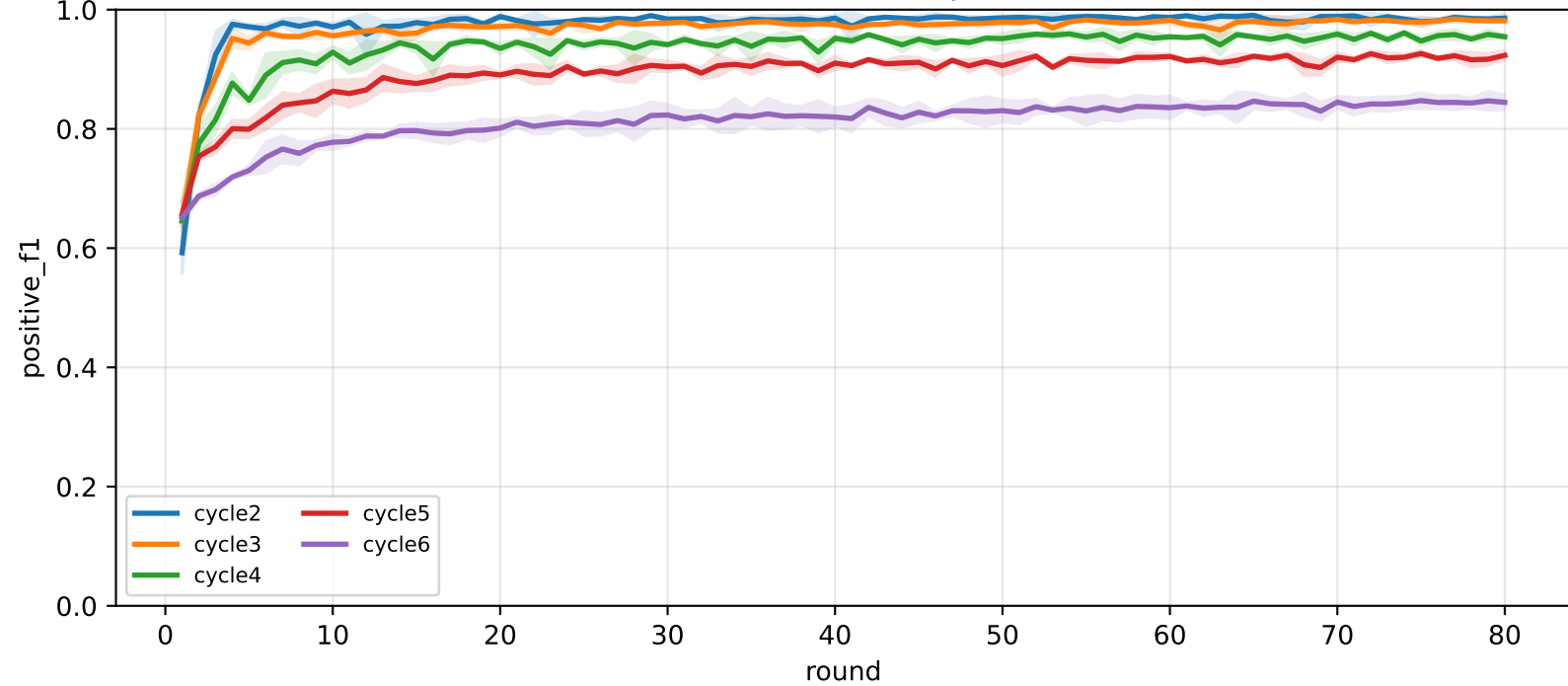
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 0.3%	93.5% ± 0.2%	98.9% ± 0.3%	98.0% ± 0.1%	95.7% ± 0.7%	91.4% ± 0.5%	83.5% ± 0.9%
FedAvg	94.3% ± 0.4%	94.8% ± 0.3%	98.9% ± 0.2%	98.2% ± 0.2%	95.7% ± 0.4%	93.4% ± 0.8%	88.0% ± 0.5%
FedProx	93.1% ± 0.2%	93.9% ± 0.2%	98.9% ± 0.2%	98.3% ± 0.2%	96.0% ± 0.3%	92.4% ± 0.5%	83.9% ± 0.8%
GCFL+	93.9% ± 0.6%	94.5% ± 0.5%	98.9% ± 0.1%	98.3% ± 0.1%	95.1% ± 0.3%	93.1% ± 1.0%	87.2% ± 1.2%

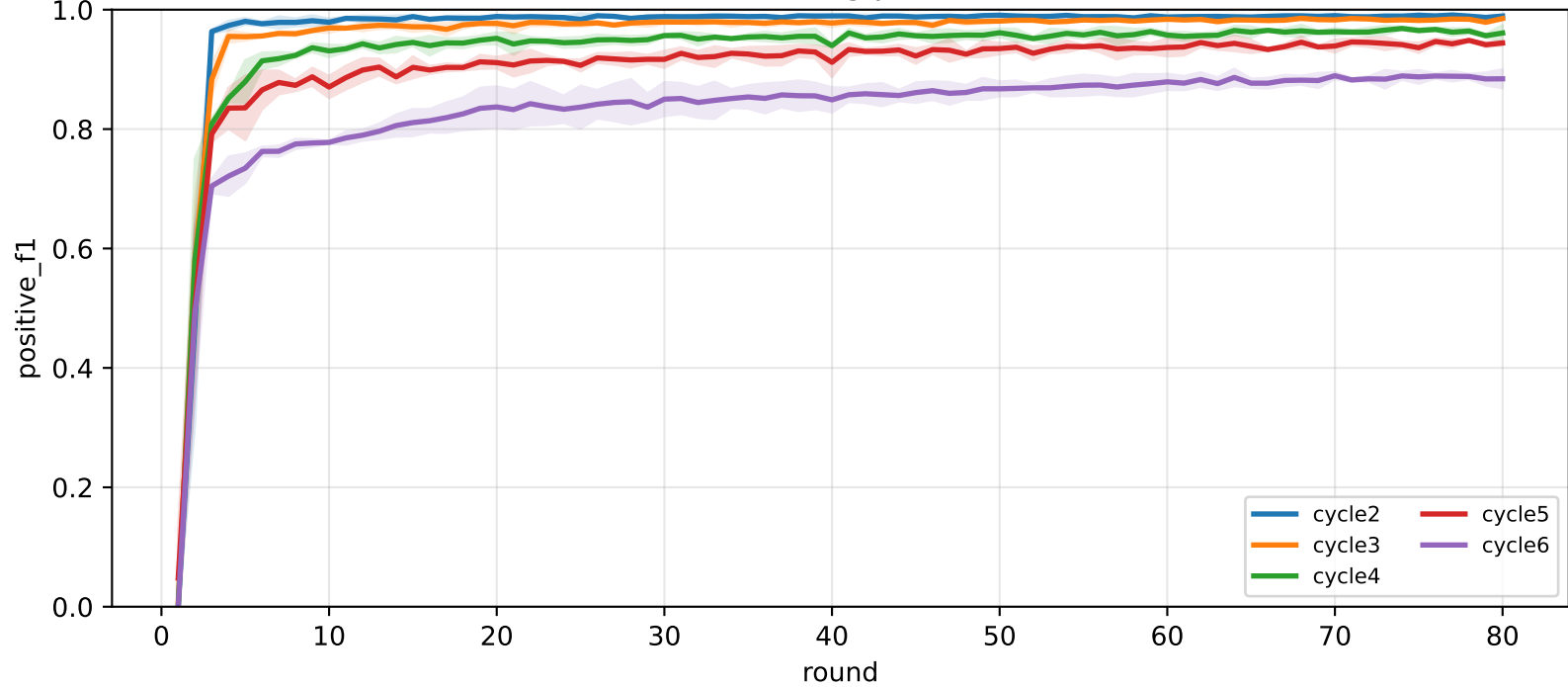
Validation loss comparison



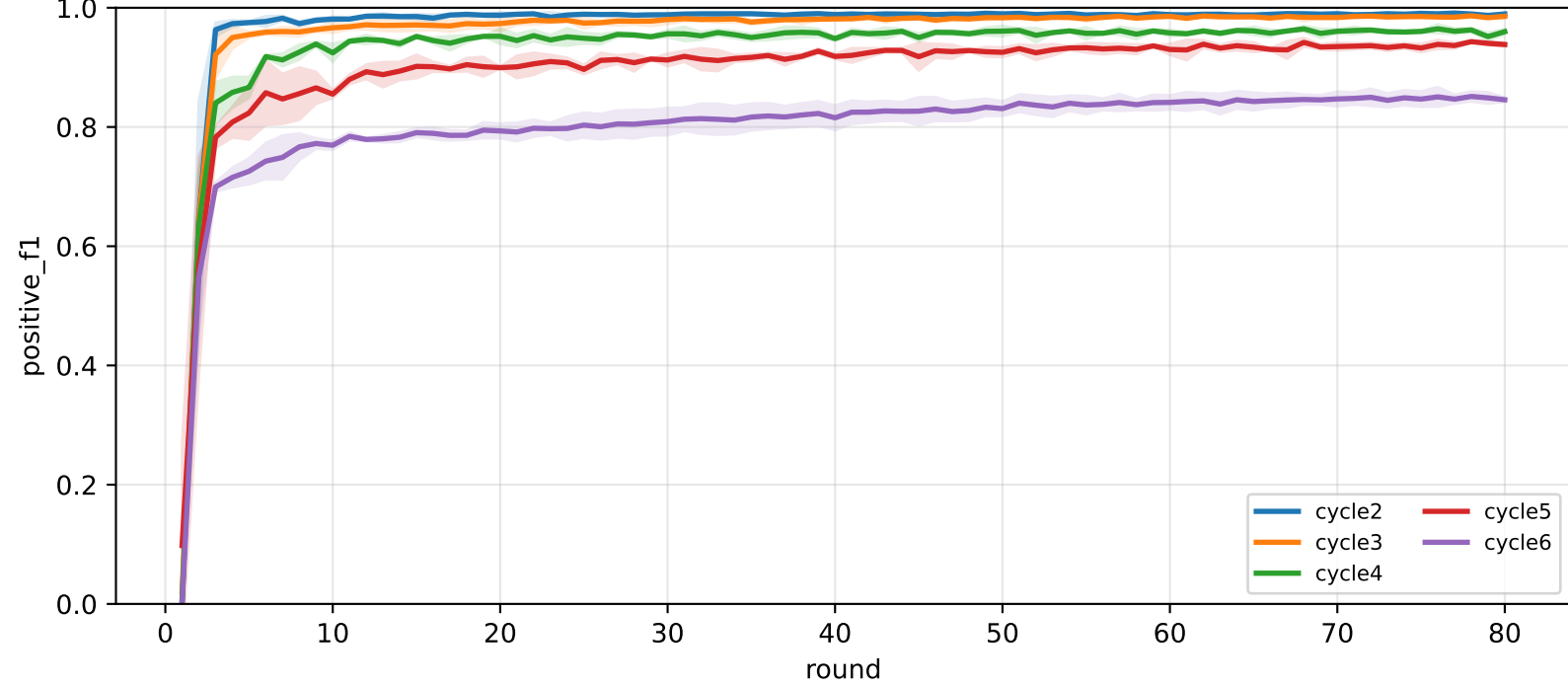
Pooled local centralized positive F1



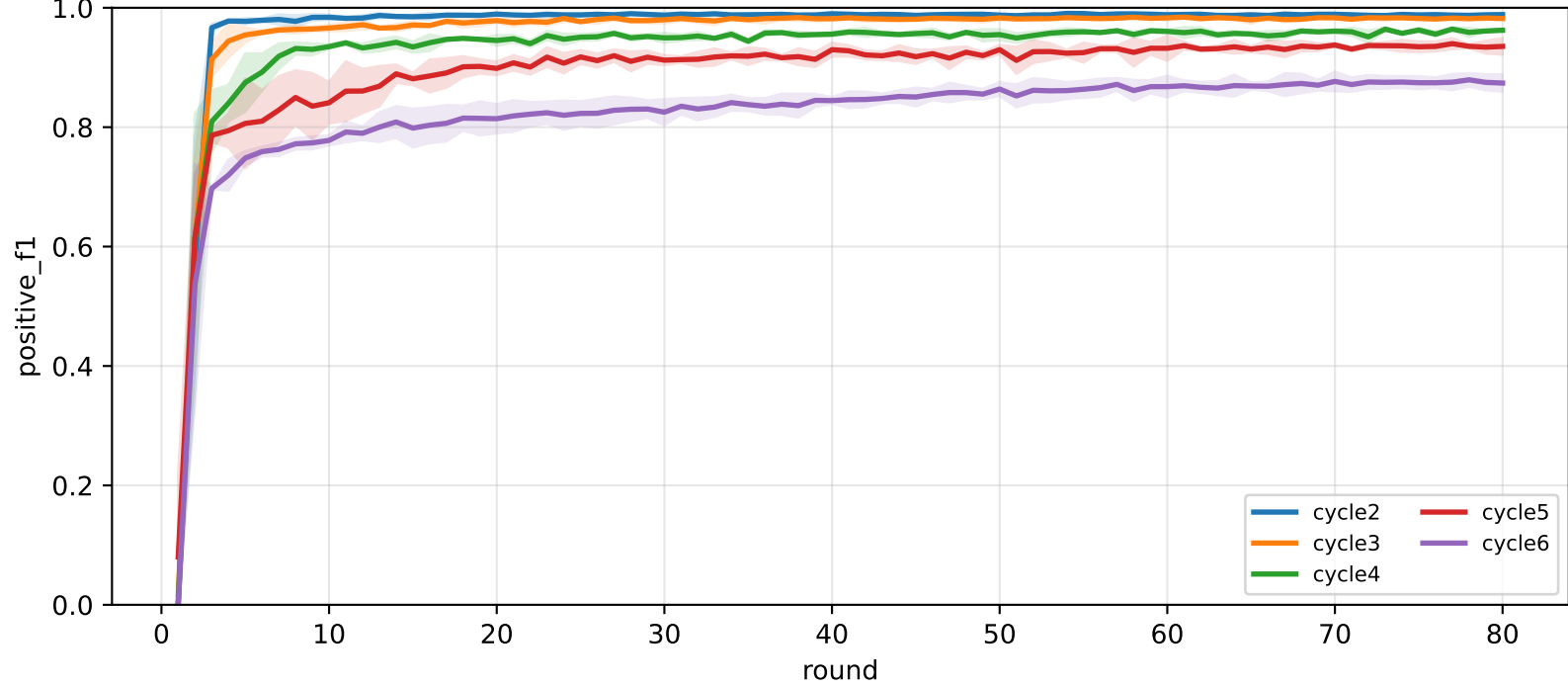
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1

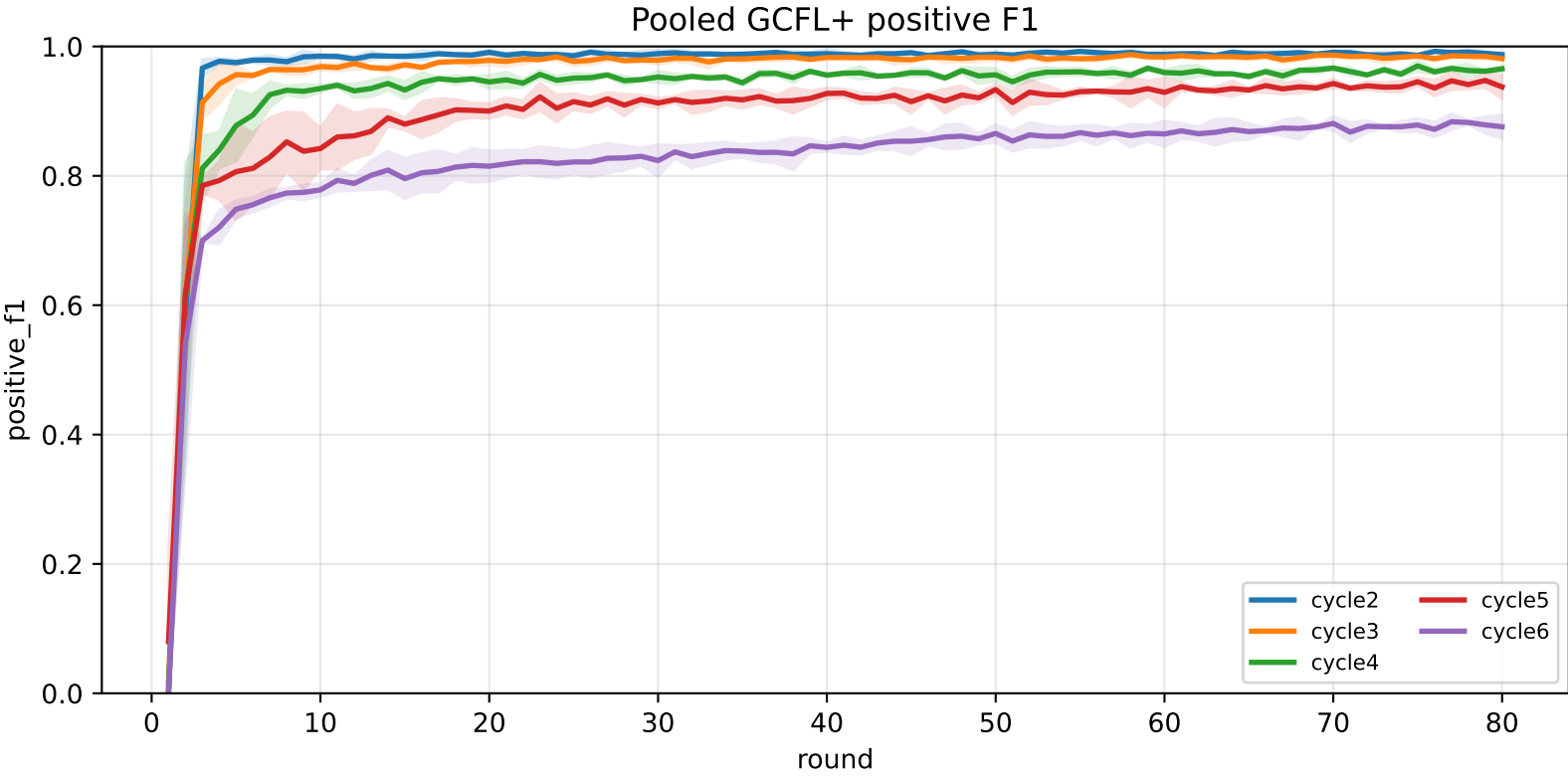
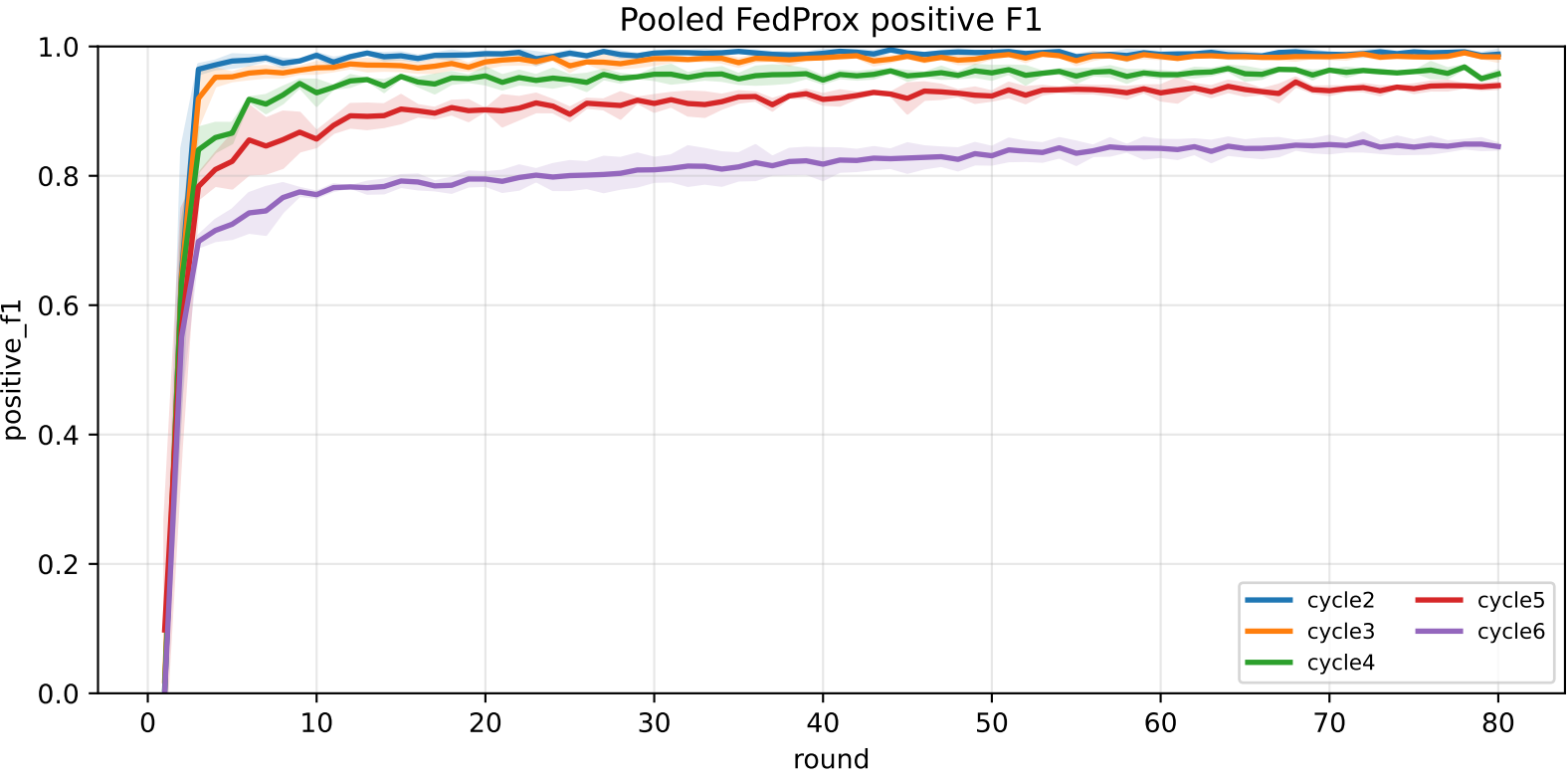
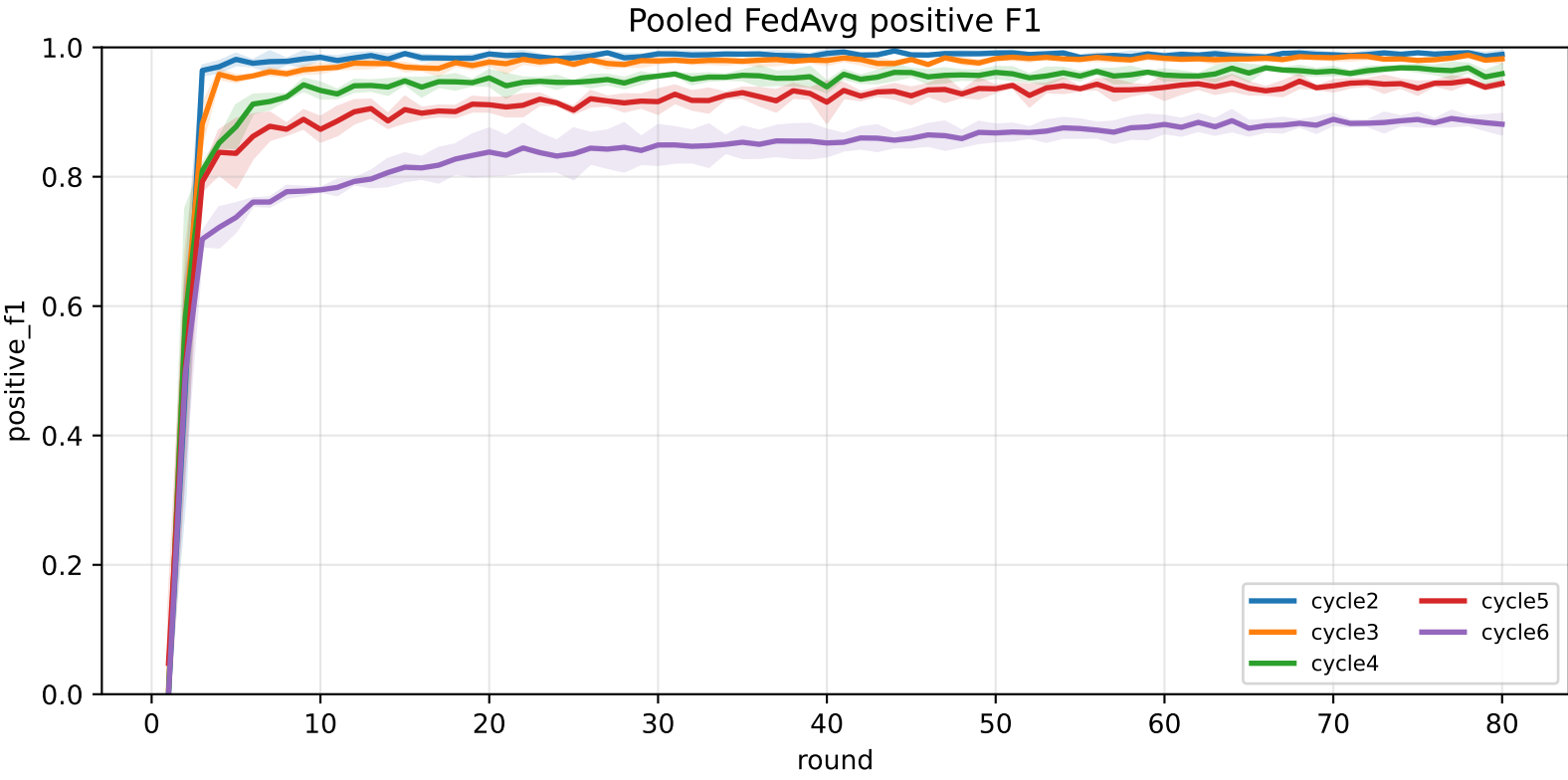
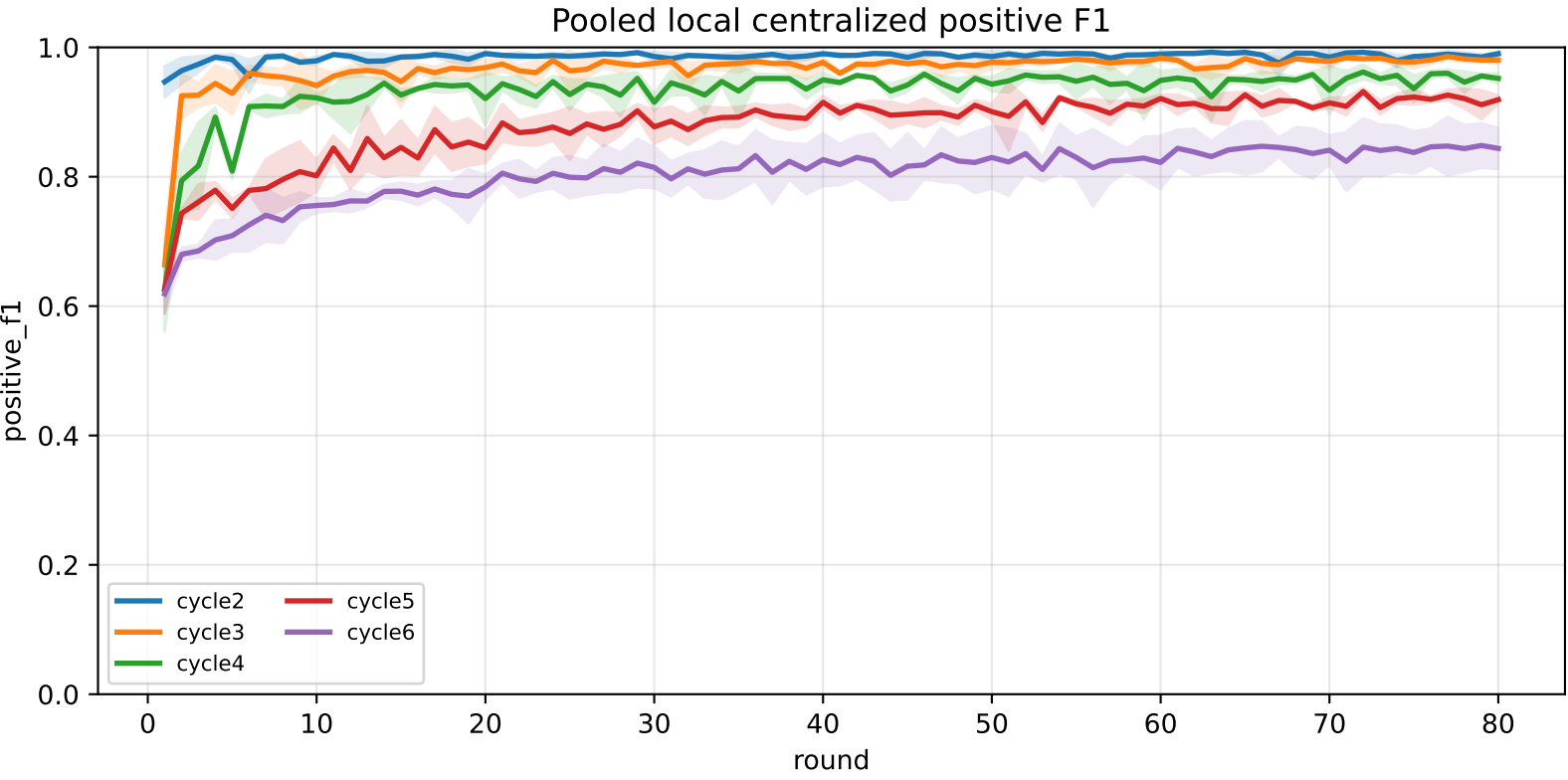
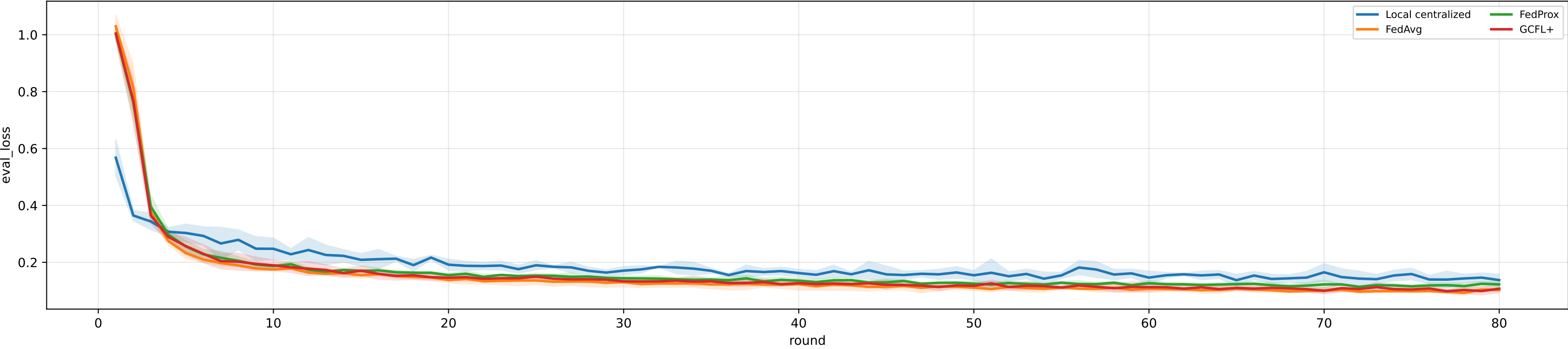


Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 1.0%	93.5% ± 0.8%	99.1% ± 0.2%	98.1% ± 0.1%	95.2% ± 0.9%	91.2% ± 1.4%	84.0% ± 3.0%
FedAvg	94.5% ± 0.4%	95.0% ± 0.4%	99.1% ± 0.4%	98.4% ± 0.3%	95.9% ± 0.3%	93.6% ± 1.0%	88.1% ± 0.3%
FedProx	93.3% ± 0.3%	94.1% ± 0.3%	99.1% ± 0.4%	98.4% ± 0.3%	96.2% ± 0.3%	92.5% ± 0.7%	84.2% ± 1.1%
GCFL+	93.9% ± 0.8%	94.5% ± 0.7%	98.9% ± 0.4%	98.2% ± 0.3%	94.9% ± 0.3%	93.2% ± 1.3%	87.0% ± 1.4%

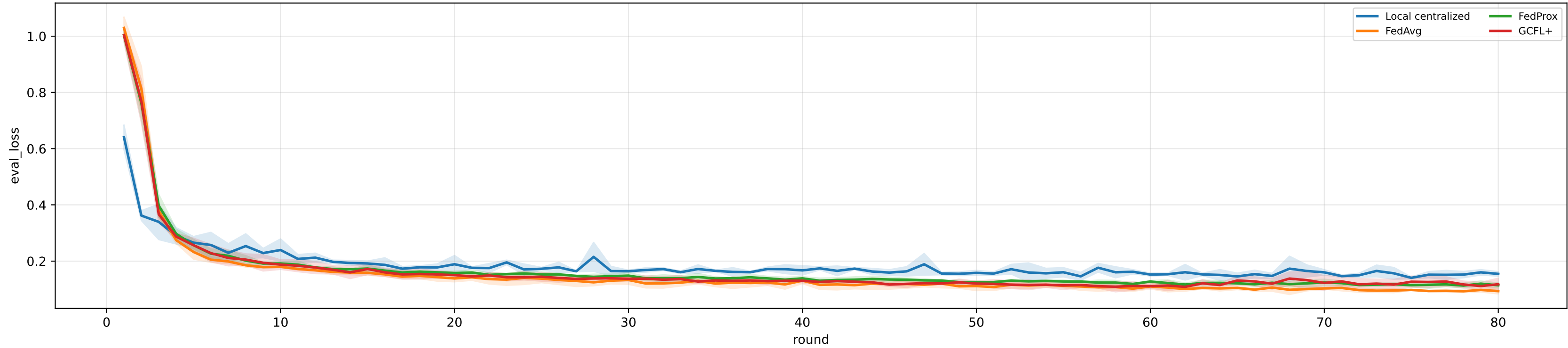
Validation loss comparison



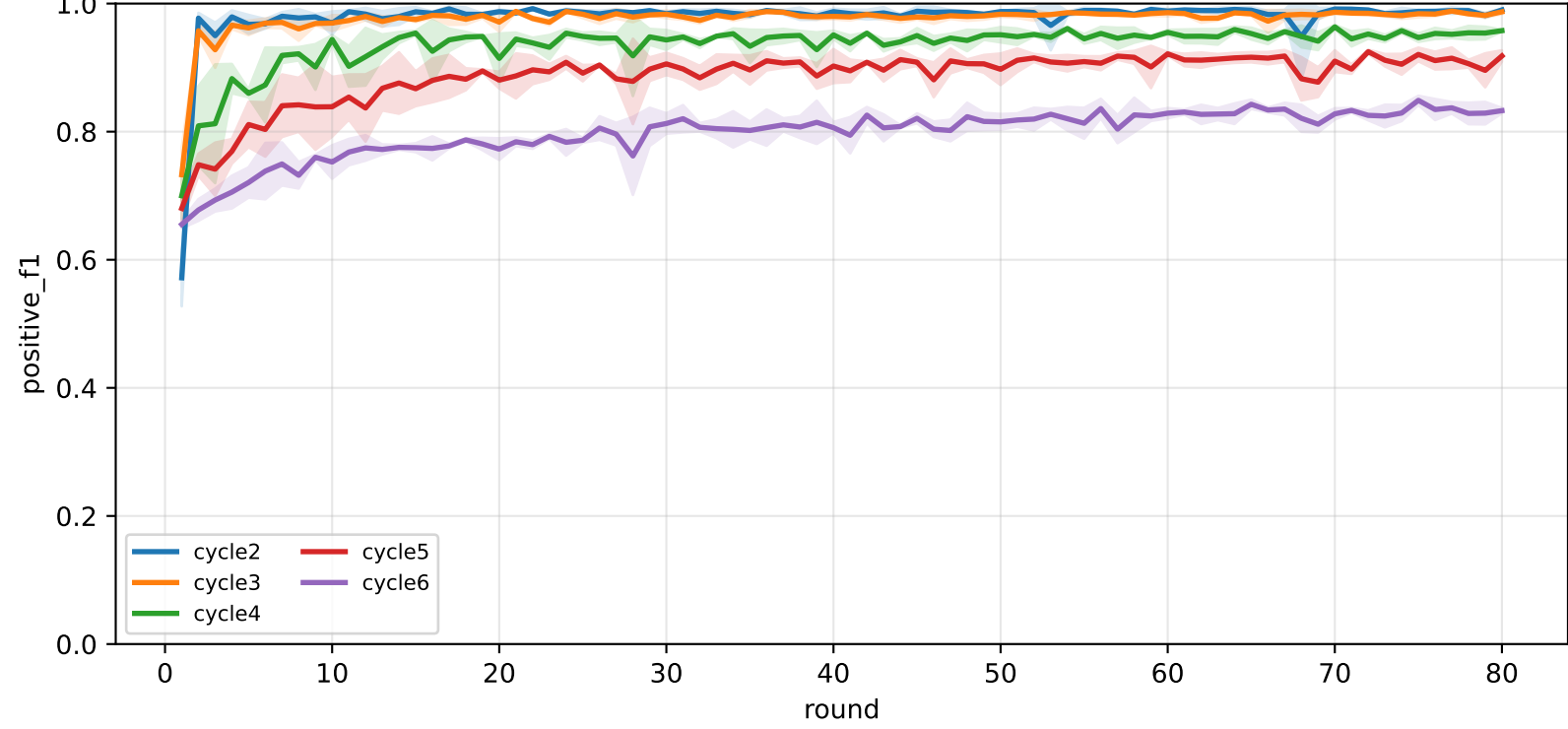
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.7% ± 0.2%	92.7% ± 0.2%	98.9% ± 0.1%	97.9% ± 0.1%	94.7% ± 1.2%	90.1% ± 0.6%	81.9% ± 0.3%
FedAvg	94.0% ± 0.5%	94.6% ± 0.4%	98.8% ± 0.2%	97.6% ± 0.3%	95.4% ± 0.8%	93.3% ± 0.8%	87.7% ± 0.5%
FedProx	93.0% ± 0.2%	93.7% ± 0.2%	98.9% ± 0.3%	97.9% ± 0.4%	95.7% ± 0.5%	92.6% ± 0.5%	83.6% ± 0.9%
GCFL+	93.5% ± 0.7%	94.1% ± 0.6%	98.6% ± 0.5%	98.5% ± 0.1%	94.4% ± 0.6%	92.8% ± 1.3%	86.3% ± 1.0%

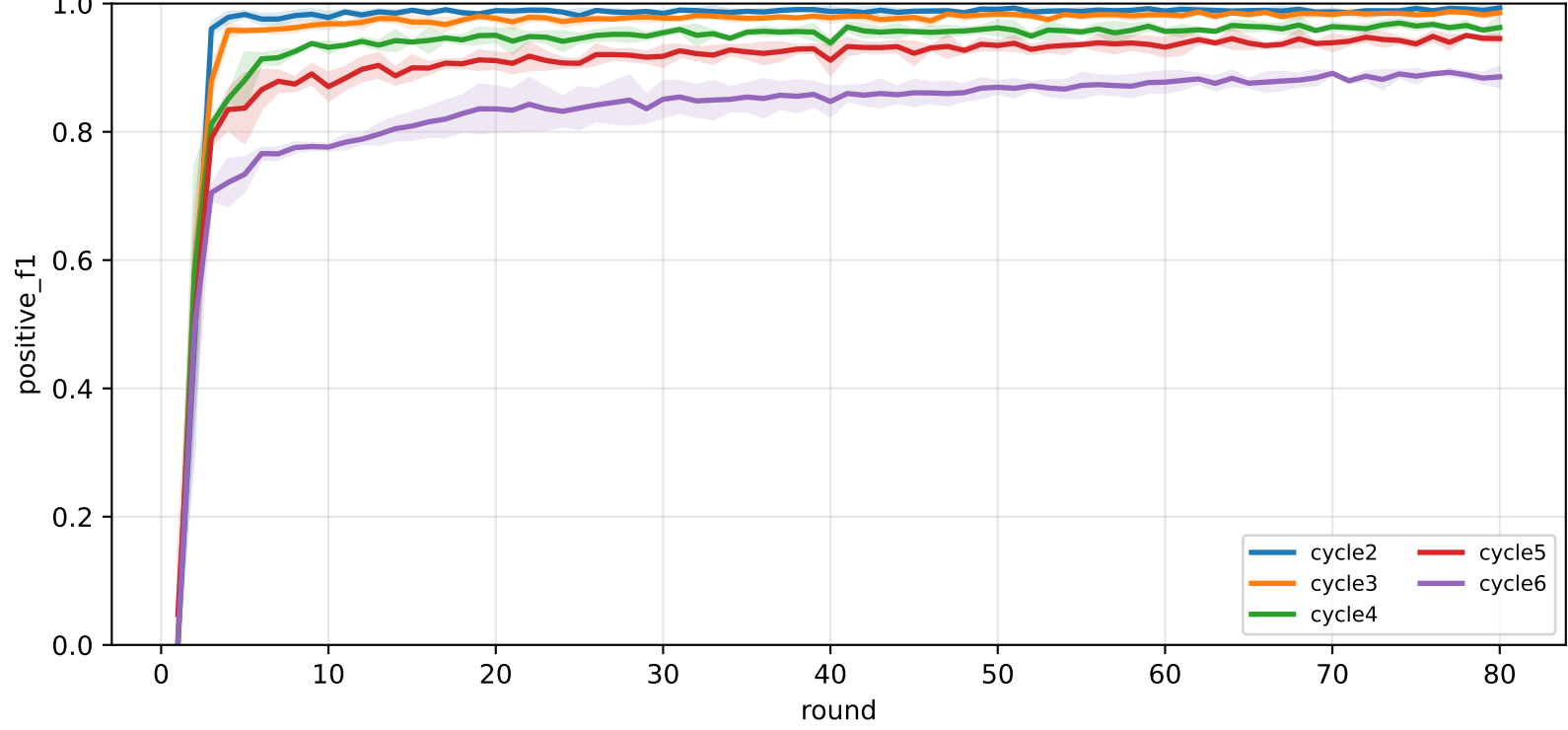
Validation loss comparison



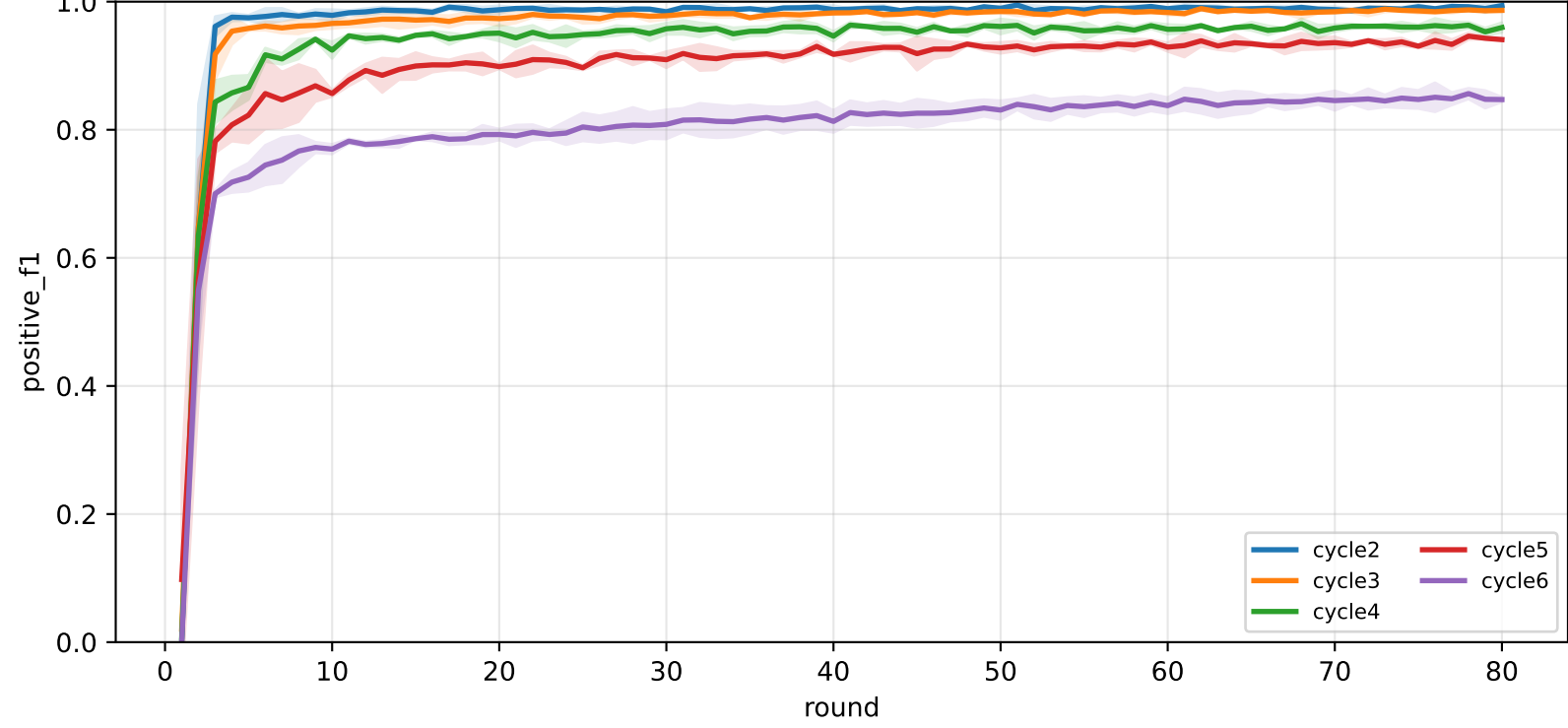
Pooled local centralized positive F1



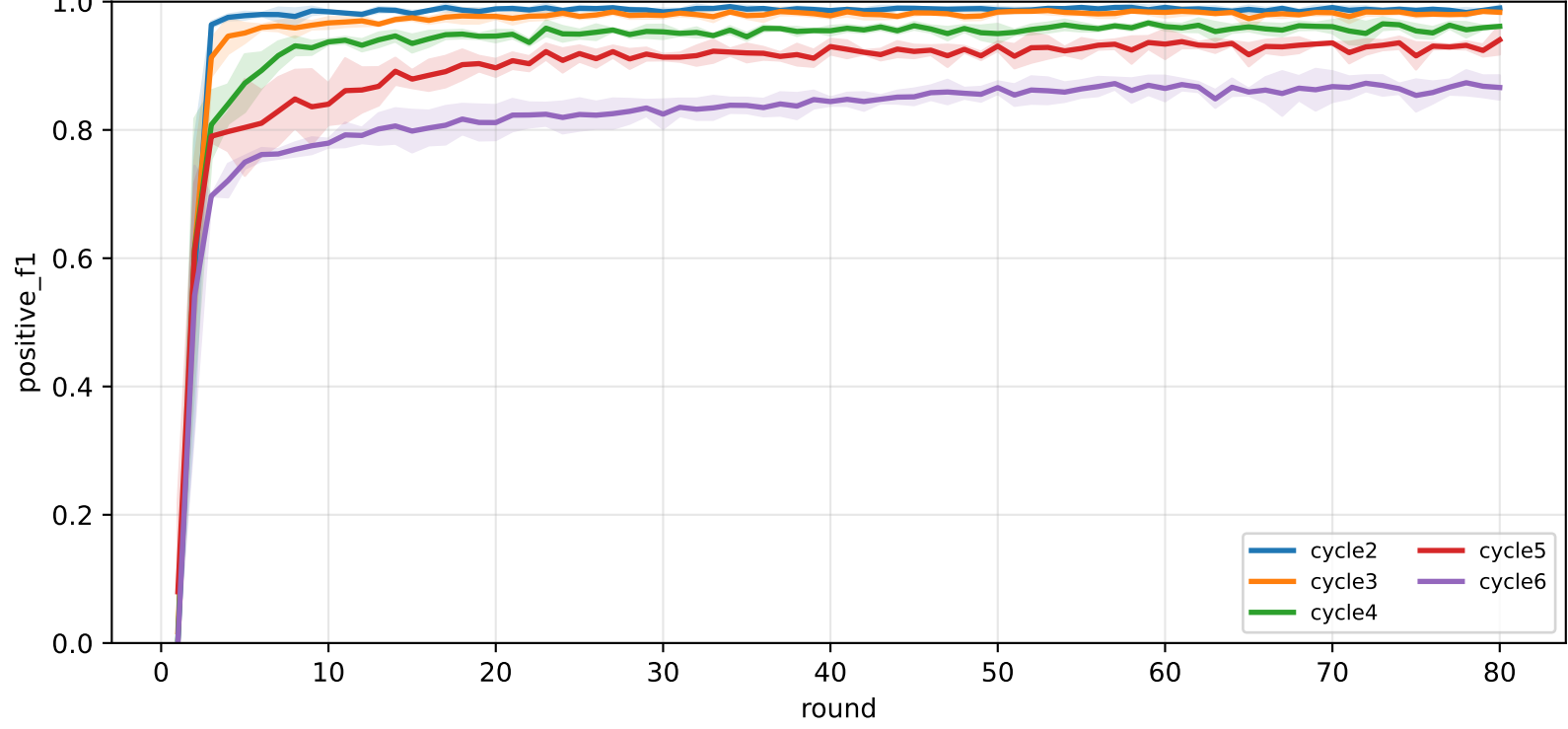
Pooled FedAvg positive F1



Pooled FedProx positive F1

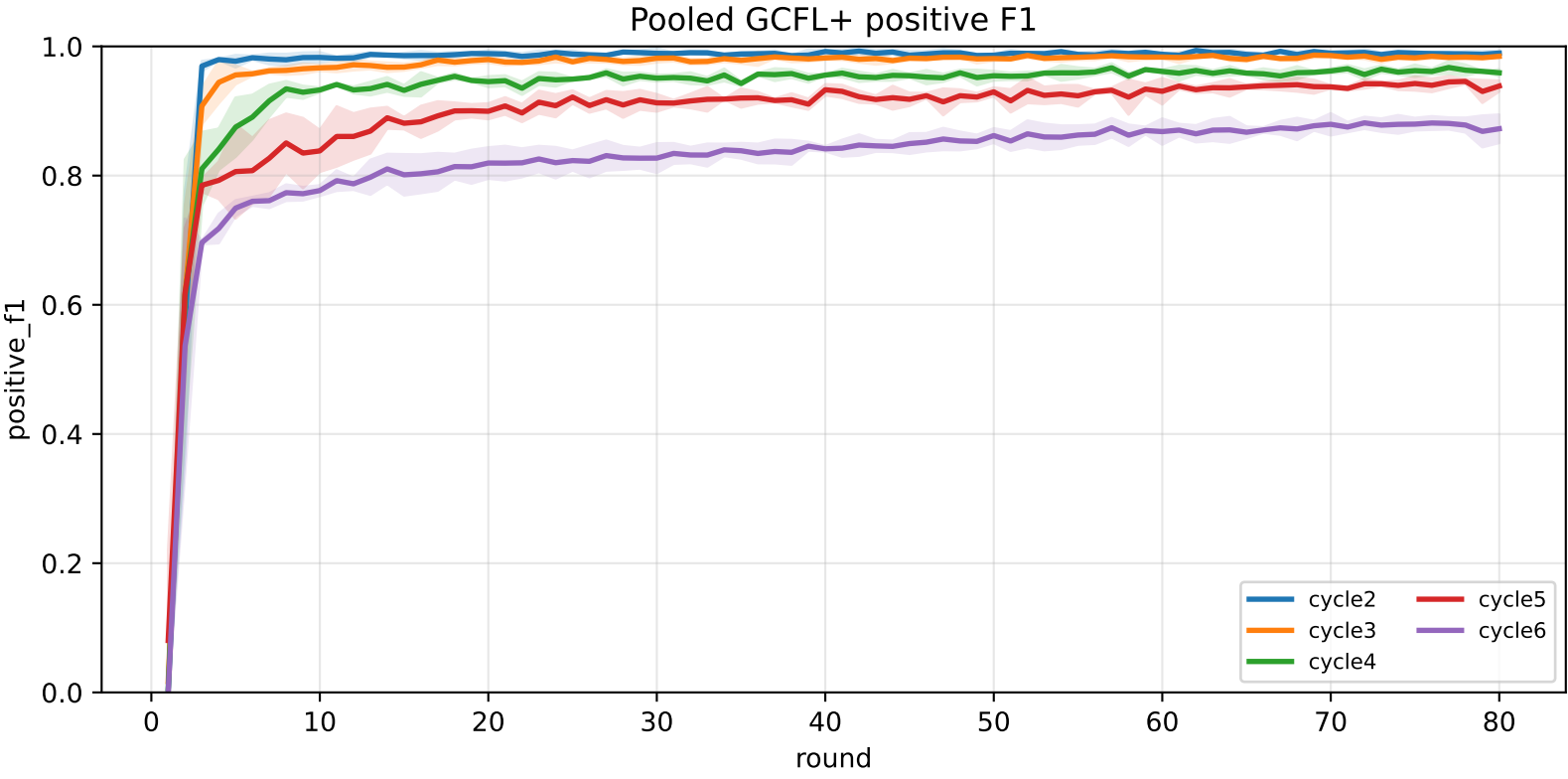
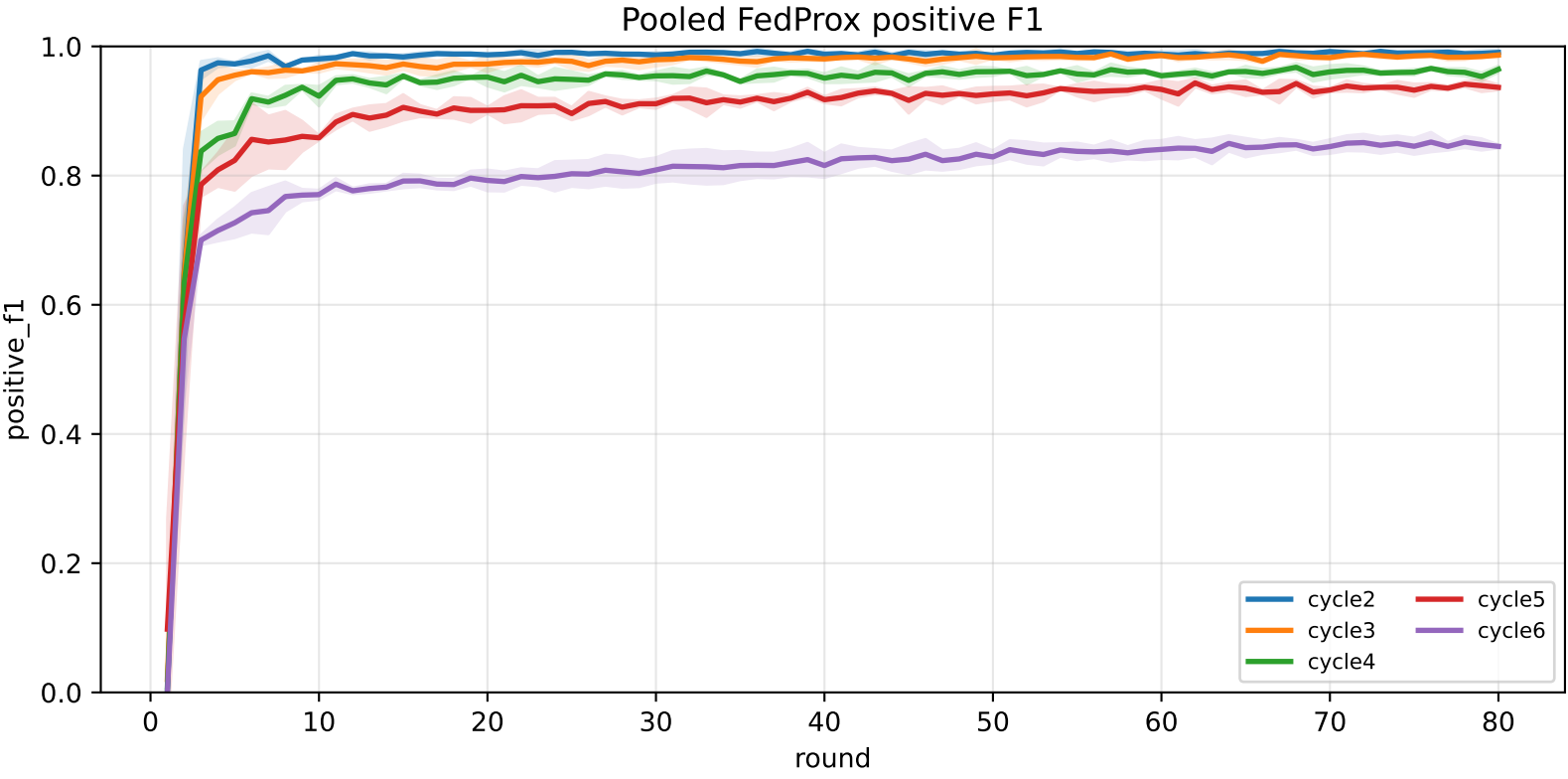
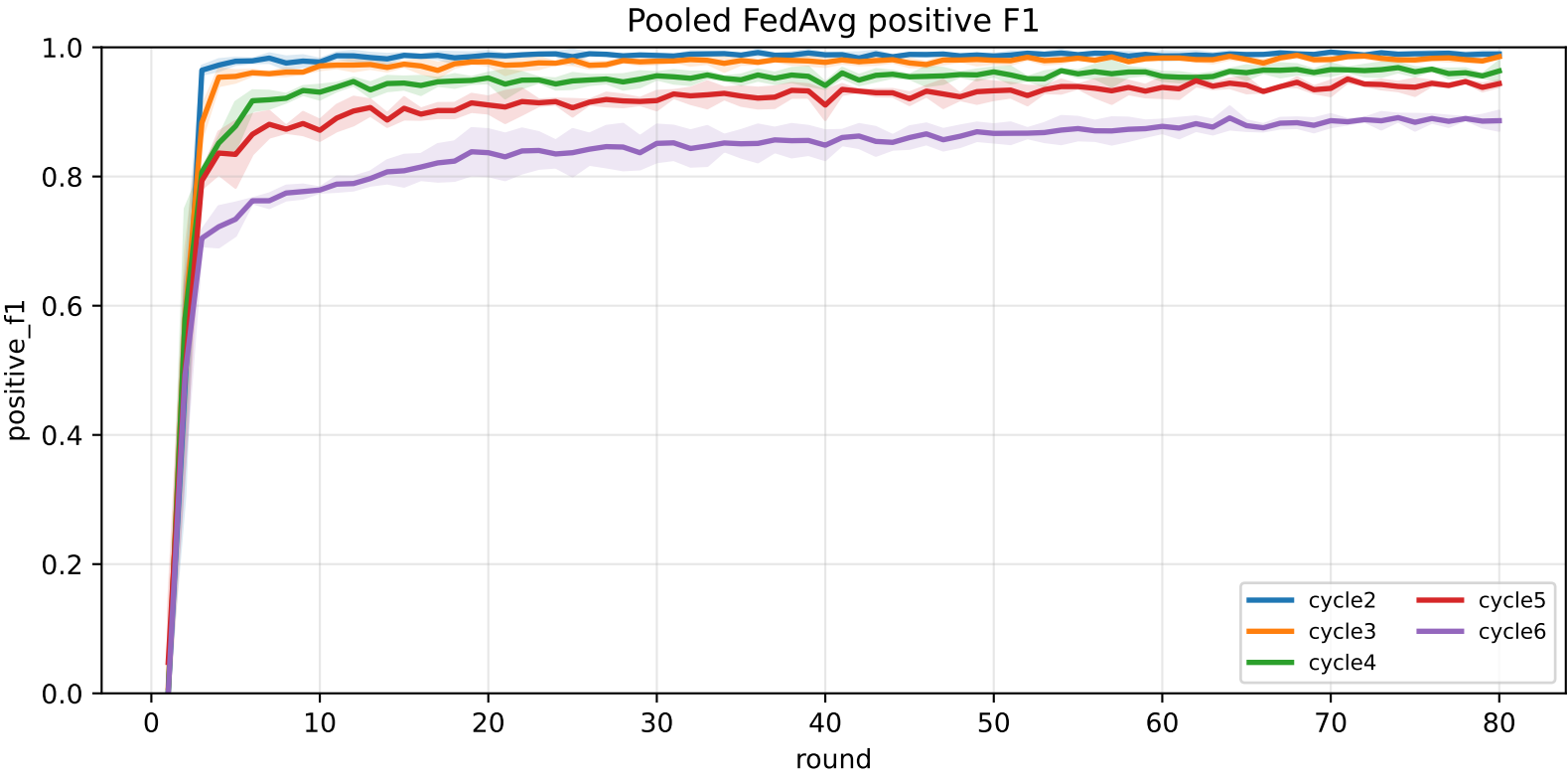
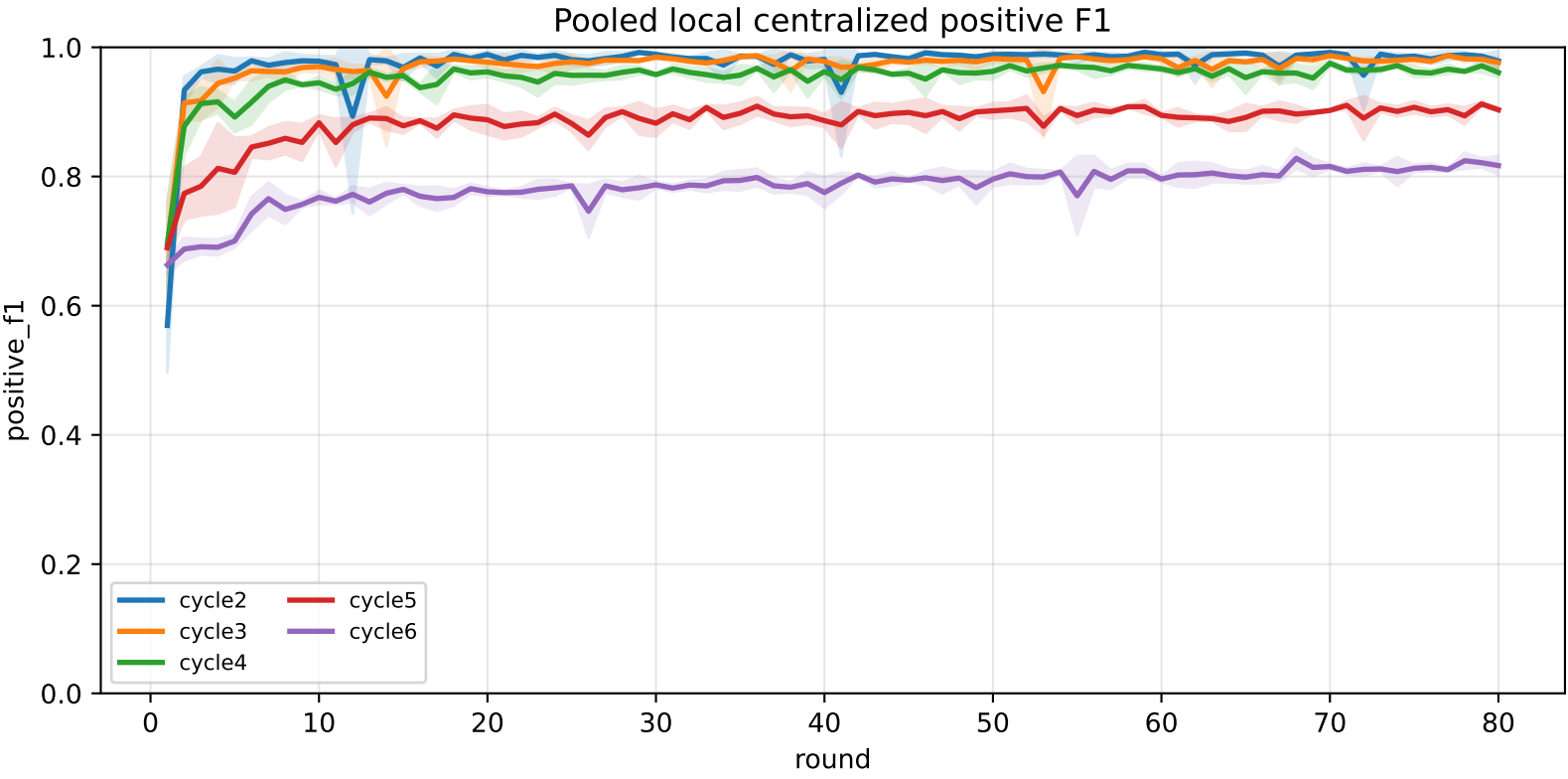
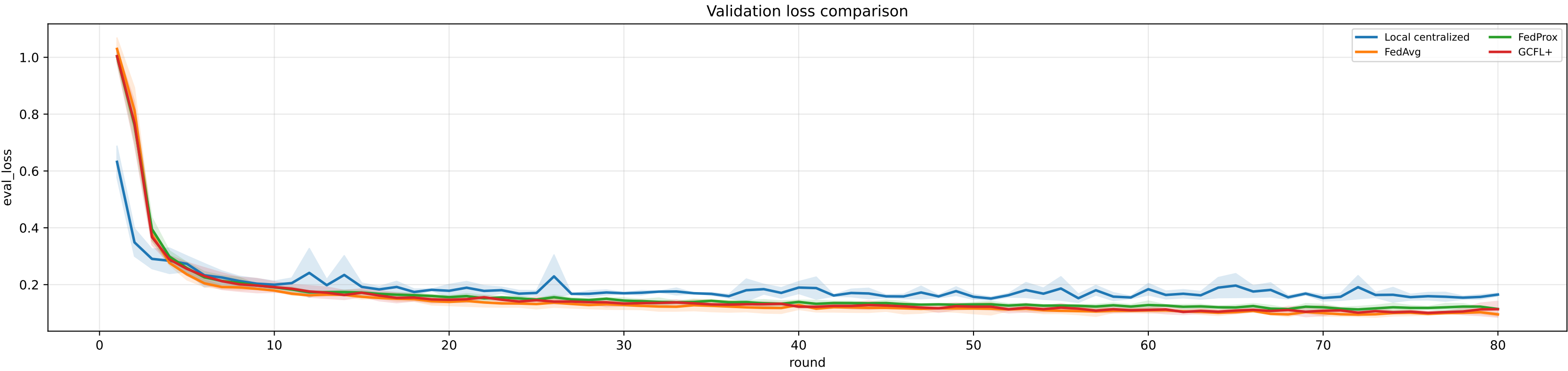


Pooled GCFL+ positive F1



family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.5% ± 0.4%	92.6% ± 0.2%	98.9% ± 0.4%	98.2% ± 0.1%	96.7% ± 0.5%	89.2% ± 0.2%	80.2% ± 1.5%
FedAvg	94.3% ± 0.3%	94.9% ± 0.3%	99.1% ± 0.2%	98.1% ± 0.2%	96.0% ± 0.5%	93.1% ± 0.5%	88.0% ± 0.4%
FedProx	93.0% ± 0.3%	93.8% ± 0.2%	99.1% ± 0.2%	98.2% ± 0.2%	96.0% ± 0.4%	92.1% ± 0.5%	83.7% ± 1.1%
GCFL+	94.1% ± 0.8%	94.7% ± 0.6%	99.2% ± 0.2%	98.4% ± 0.4%	95.3% ± 0.2%	93.5% ± 1.4%	87.2% ± 1.6%



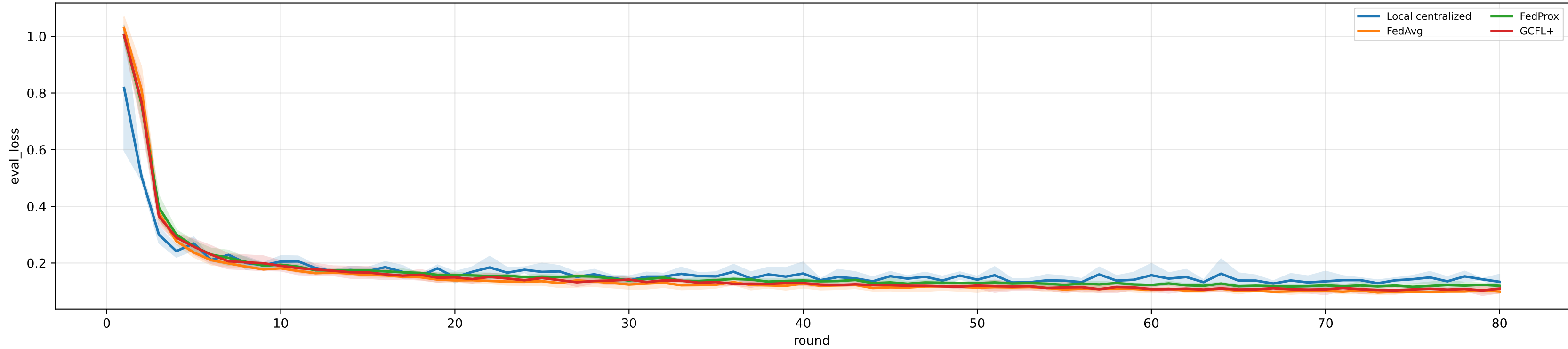
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

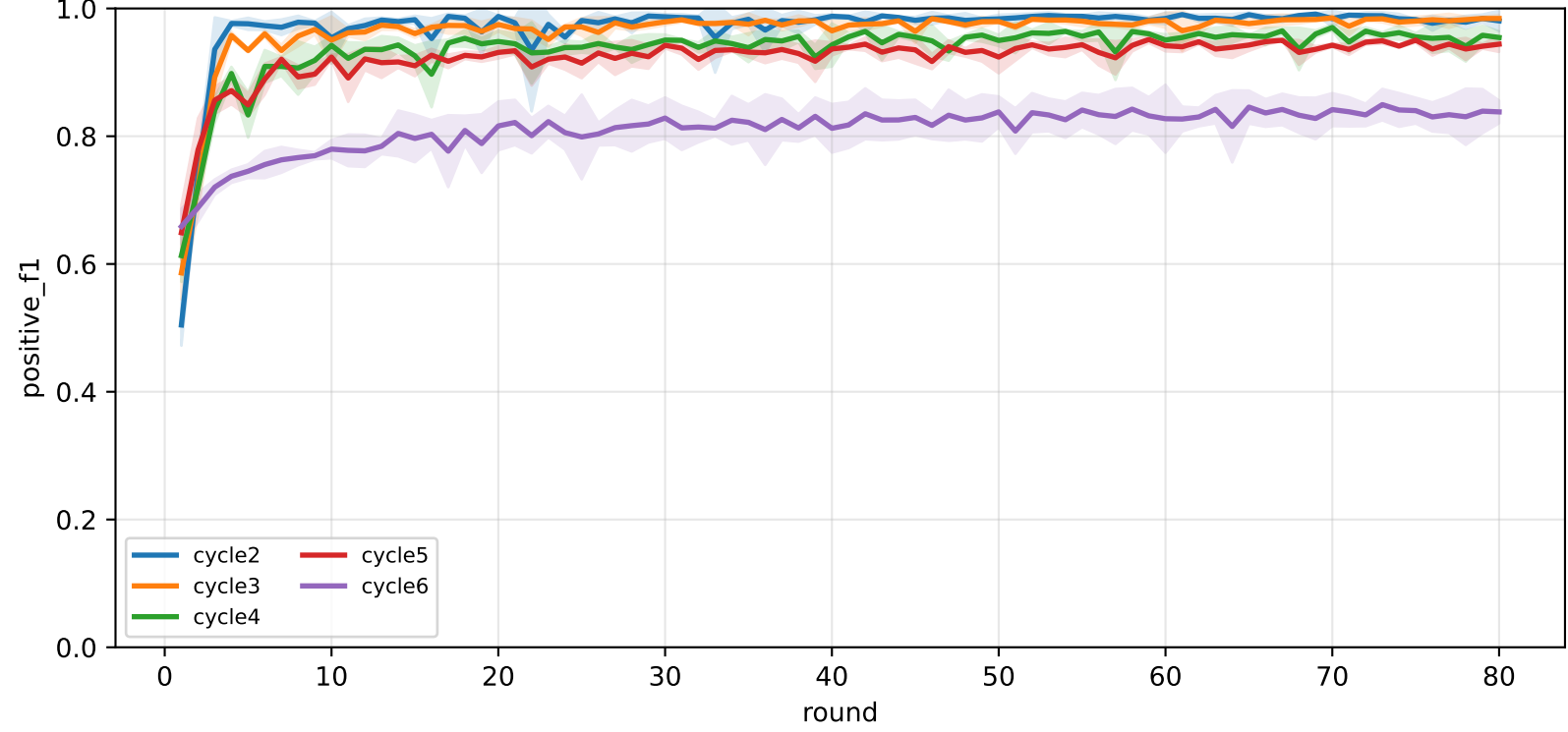
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.2% ± 0.7%	94.0% ± 0.6%	98.6% ± 0.5%	97.9% ± 0.1%	96.2% ± 0.5%	94.4% ± 0.3%	82.9% ± 2.4%
FedAvg	94.3% ± 0.4%	94.9% ± 0.3%	98.8% ± 0.4%	98.6% ± 0.0%	95.6% ± 0.6%	93.6% ± 0.9%	88.0% ± 0.6%
FedProx	93.2% ± 0.2%	94.0% ± 0.2%	98.8% ± 0.3%	98.7% ± 0.2%	96.0% ± 0.4%	92.5% ± 0.5%	84.0% ± 0.7%
GCFL+	94.1% ± 0.8%	94.7% ± 0.6%	99.0% ± 0.1%	98.5% ± 0.2%	95.3% ± 0.8%	93.3% ± 1.1%	87.6% ± 1.2%

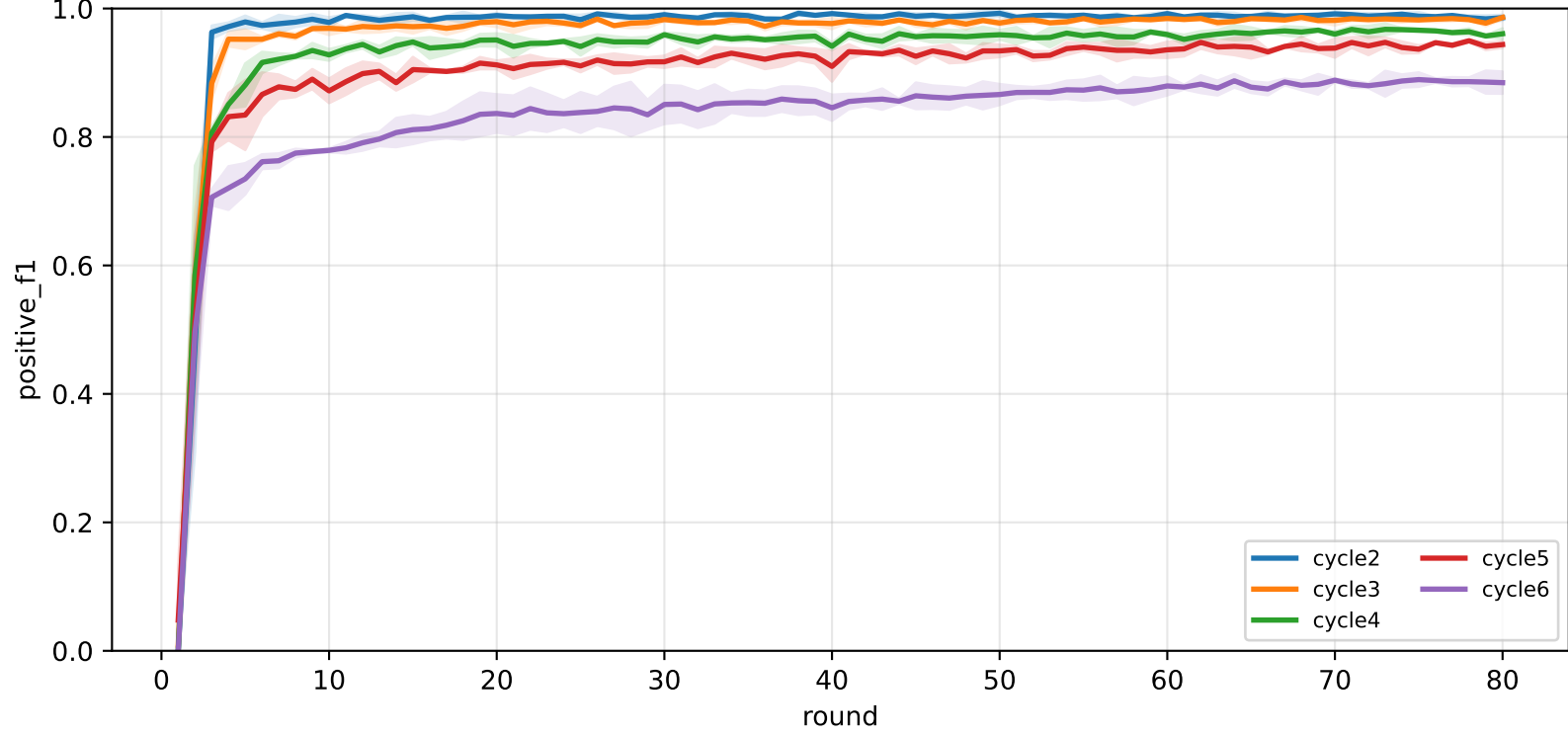
Validation loss comparison



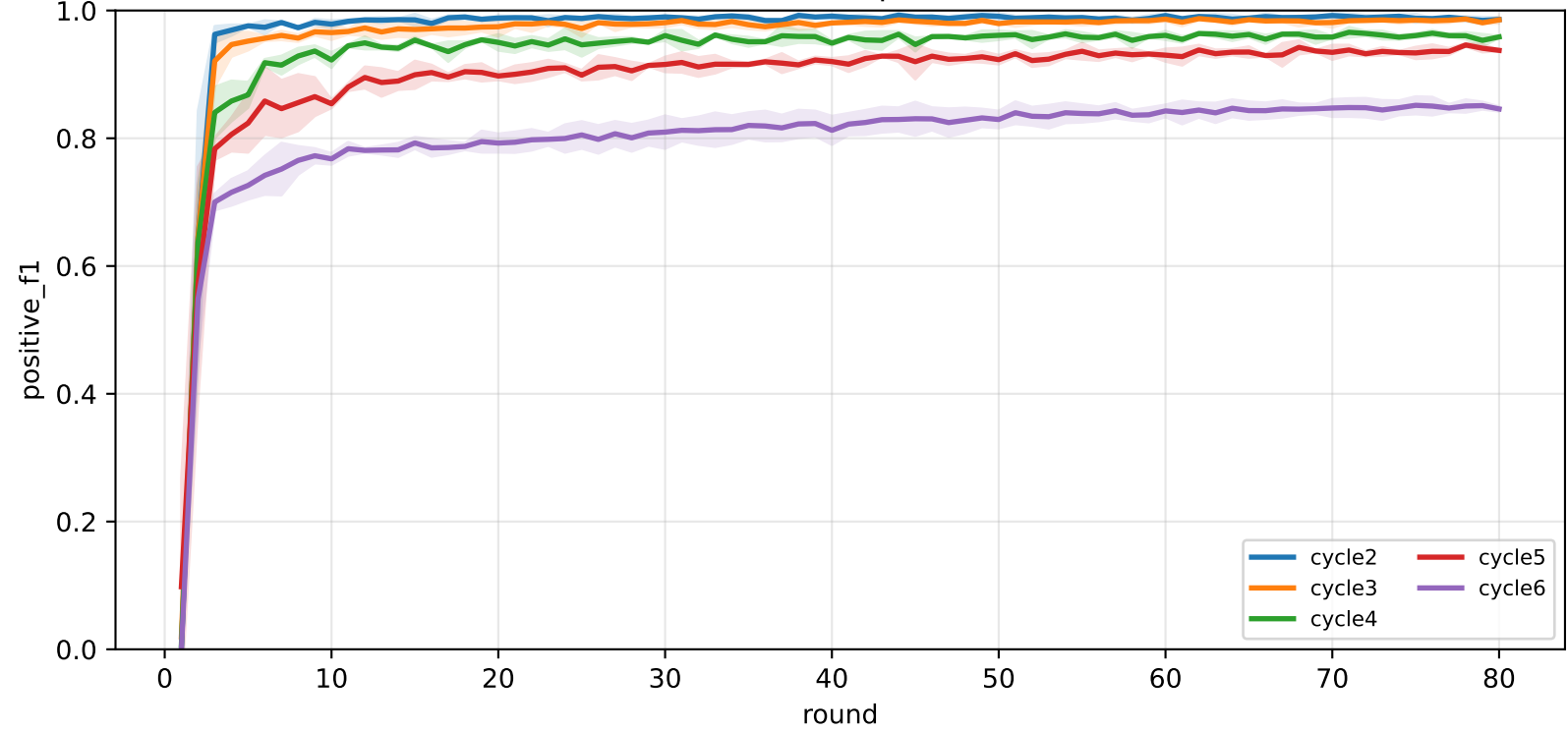
Pooled local centralized positive F1



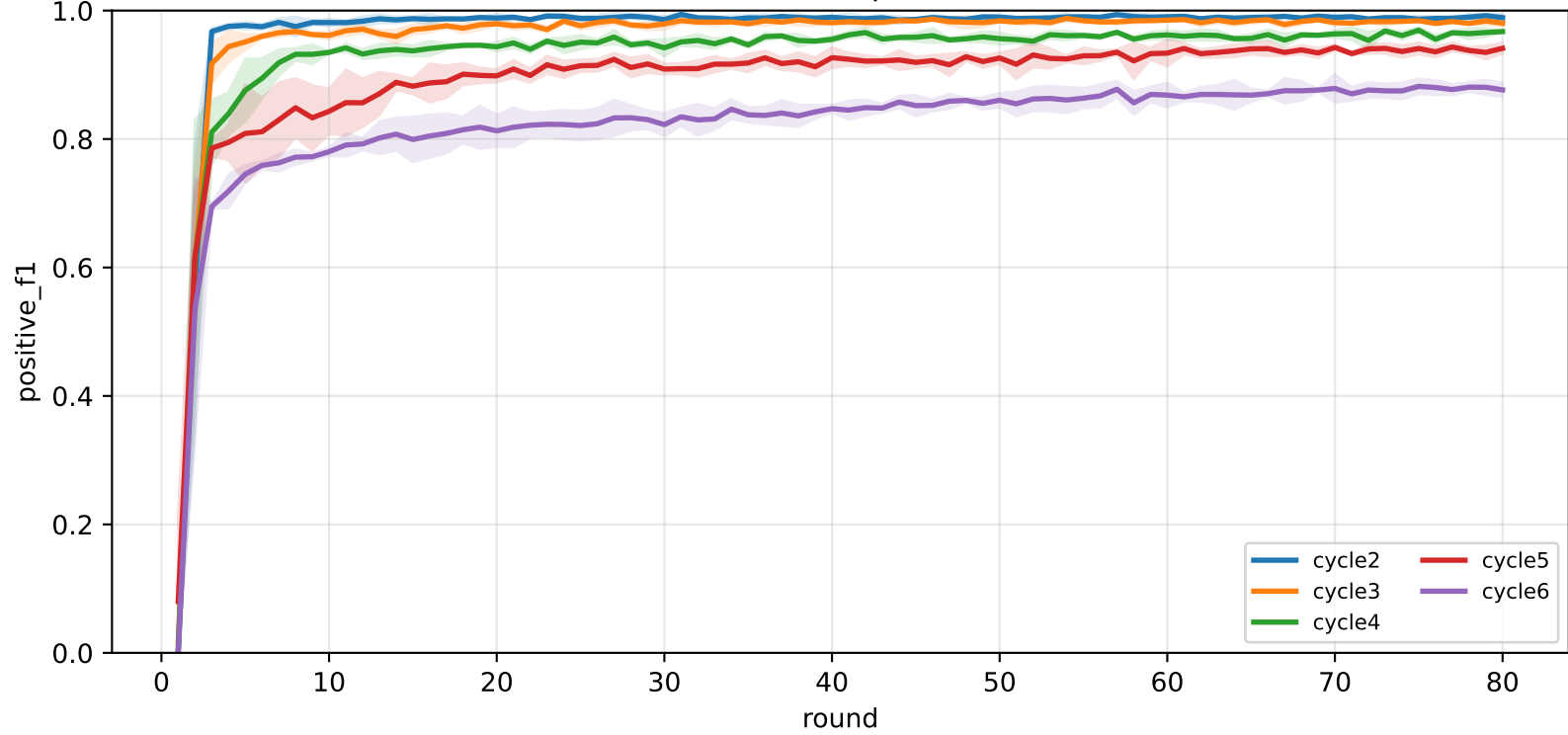
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.5%	94.7% ± 0.4%	98.9% ± 0.3%	97.9% ± 0.3%	95.5% ± 0.4%	92.4% ± 1.0%	88.7% ± 1.2%
FedAvg	94.3% ± 0.4%	94.9% ± 0.3%	98.8% ± 0.4%	98.3% ± 0.4%	95.7% ± 0.4%	93.3% ± 0.8%	88.2% ± 0.5%
FedProx	93.2% ± 0.2%	93.9% ± 0.1%	98.9% ± 0.4%	98.2% ± 0.4%	95.9% ± 0.2%	92.4% ± 0.5%	84.2% ± 0.3%
GCFL+	94.1% ± 0.6%	94.7% ± 0.6%	98.9% ± 0.4%	98.1% ± 0.1%	95.7% ± 0.5%	92.8% ± 1.6%	87.9% ± 1.2%

